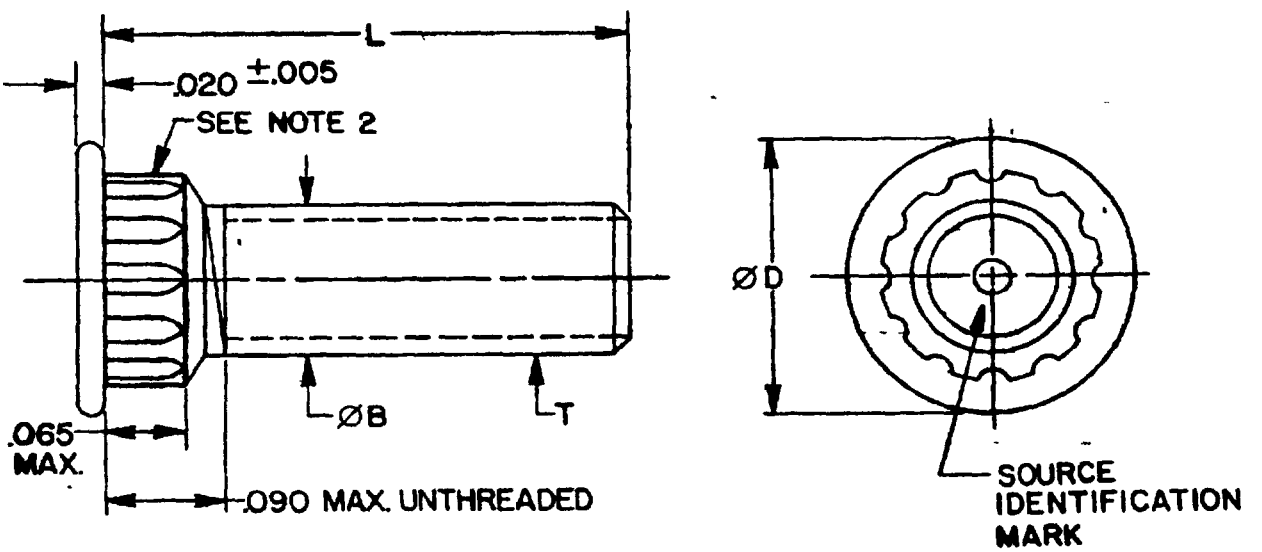
MILITARY SPECIFICATION SHEET

STUD, SELF-LOCKING, BROACHING TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-S-63540



NOTES:

1. All dimensions are in inches.
2. Broaching profile may vary.

FIGURE 1. STUD, SELF-LOCKING, BROACHING TYPE

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. PART DIMENSIONS

T THREAD SIZE CLASS 2A	ØB STUD	ØD ±.010
.112-40UNC (4-40)	.112	.180
.138-32UNC (6-32)	.138	.200
.164-32UNC (8-32)	.164	.225
.190-32UNF (10-32)	.190	.250

TABLE II. THREAD SIZE AND LENGTH DASH NUMBERS

L ±.010	T THREAD SIZE CLASS 2A			
	.112-40 UNC	.138-32 UNC	.164-32 UNC	.190-32UNF
	THREAD SIZE AND LENGTH DASH NUMBER			
.250	1	7		
.312	2	8	13	
.375	3	9	14	18
.500	4	10	15	19
.625	5	11	16	20
.750	6	12	17	21

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REQUIREMENTS:

1. Material. Phosphor bronze shall be Copper Alloy UNS C51000 in accordance with ASTM B159 , temper H01.

2. Protective finish code.

"T" = Electrodeposited tin

"L" = Electrodeposited tin-lead

3. Hardness. Phosphor bronze studs shall have a hardness of 121HB minimum (68HRB minimum).

4. Push-out and torque-out values. When the stud is properly installed, (see installation procedures), push-out and torque-out values shall be as specified in table III.

TABLE III. PUSH-OUT AND TORQUE-OUT VALUES.

T THREAD SIZE CLASS 2A	BOARD THICK-NESS MIN (REF)	PUSH-OUT LOAD (LBS) MIN 1/	TORQUE-OUT LOAD (IN LBS) MIN 1/
		G-10 FIBERGLASS SHEET MATERIAL	
.112-40 UNC	.060	65	7
.138-32 UNC		70	11
.164-32 UNC		80	16
.190-32 UNF		90	17

1/ The load values shown are the minimum non-destruct values that apply during testing.

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5. Part number. The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table II, followed by a protective finish code letter.

Example: M63540/7-10 T

- T — Protective finish code
- 10 — Dash number
- M63540/7 — Specification sheet number

M63540/7-10T indicates - Stud, self-locking, broaching type; Phosphor bronze with electrodeposited tin; .138-32UNC-2A thread; .500 length; .200 head diameter; .020 ±.005 head height.

M63540/7-20L indicates - Stud, self-locking, broaching type; Phosphor bronze with electrodeposited tin-lead; .190-32 UNF-2A thread; .625 length; .250 head diameter; .020 ±.005 head height.

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INSTALLATION PROCEDURES:

a. Any board thickness above the minimum indicated on table IV for a specific thread size may be used.

b. Drill hole in board material with diameter as shown in table IV.

c. Locate shank of stud squarely in the hole in the mounting surface, with anvil below board (see Figure 2).

d. Apply force until the underside of the head of stud is in contact with mounting material.

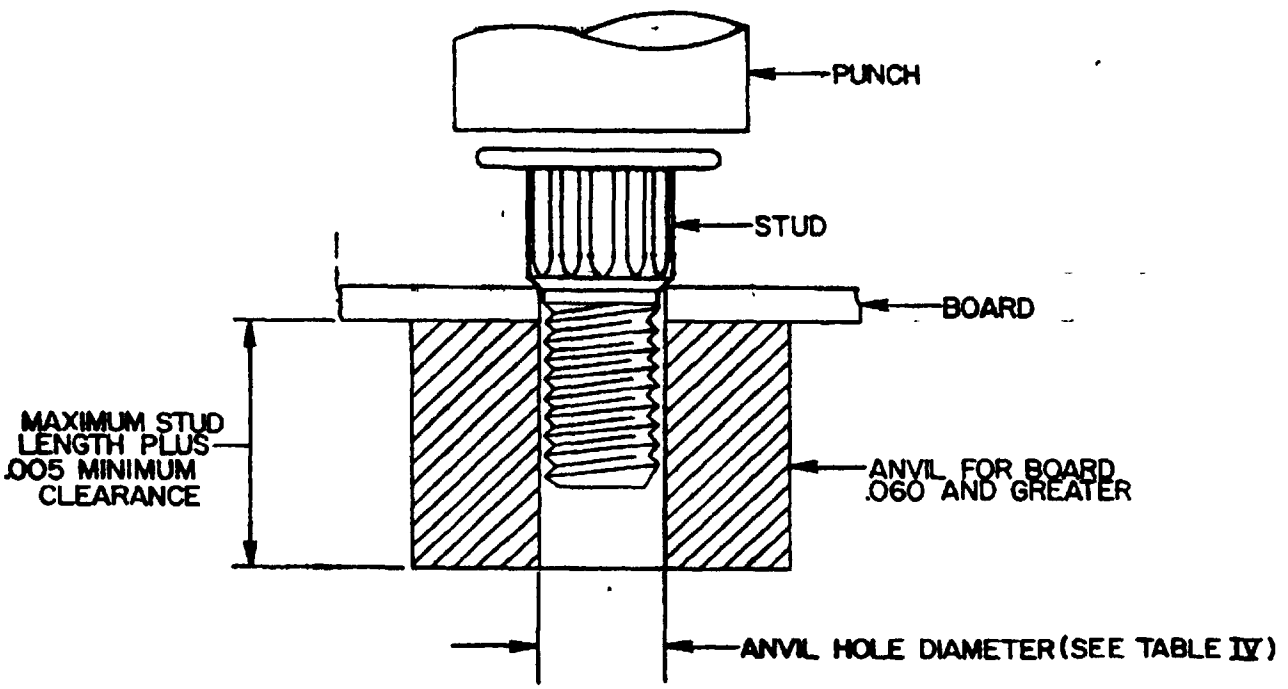


FIGURE 2. GENERAL TOOLING SETUP.

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TABLE IV. INSTALLATION DIMENSIONS - (REFERENCE)

T THREAD SIZE CLASS 2A	BOARD THICKNESS MIN	HOLE SIZE IN BOARD +.003 -.000	CENTER OF HOLE TO BOARD EDGE MIN	NUT TIGHTENING TORQUE (IN LBS) 1/ MAX	ANVIL HOLE DIAMETER +.003 -.000
.112-40 UNC (4-40)	.060	.120	.15	4	.113
.138-32 UNC (6-32)		.140	.19	8	.140
.164-32 UNC (8-32)		.166	.20	15	.166
.190-32 UNF (10-32)		.187	.20	18	.191

1/ This load is the limiting load when the nut is fastened to the stud at assembly.

Custodians:
Army - AR
Navy - MC
Air Force - 99

Review activities:
Air Force - 82
DLA - IS

User activities:
NSA - NS

Preparing activity:
Army - AR

Agent:
DLA - IS

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